MATH3303 – Lecture 26

Maximal Ideals: Criterion, R/I a Field, and Examples

Review Notes

Reference: Lee, Abstract Algebra*, Section 9.6*

Overview

Continuing from Lecture 25, this lecture develops the theory of maximal ideals.

- **The** $(x, I) = R$ **criterion** – a tool for checking maximality
- **Quotient characterisation:** I maximal $\iff$ R/I is a field
- **Every maximal ideal is prime**, but the converse fails (as witnessed by $(x) \trianglelefteq \mathbb{R}[x, y]$ and $(x) \trianglelefteq \mathbb{Z}[x]$)
- **Worked example in** $\mathbb{C}[x]$**:** for every $\alpha \in \mathbb{C}$, $(x - \alpha)$ is maximal, via the Fundamental Theorem of Algebra

Connection to Lecture 25. Lecture 25 introduced both prime and maximal ideals, but only proved the quotient test (Theorem 2.1) for prime ideals and identified the maximal ideals of $\mathbb{Z}$. The parallel quotient test for maximality is the centrepiece of this lecture and immediately yields the implication *maximal* $\Rightarrow$ *prime.*

1 The Generating-Set Criterion for Maximality

Remark 1.1: Notation (x, I) (from lecture)

For $x \in R$ and an ideal $I \trianglelefteq R$, we write

$$(x, I) := (x) + I = \{rx + i : r \in R,\, i \in I\}.$$

This is the smallest ideal containing both x and I.

Lemma 1.1: $(x, I) = R$ criterion (from lecture; cf. Lee, Thm. 9.20)

Let R be a commutative ring and I a proper ideal of R. Then

$$I \text{ is maximal} \iff (x, I) = R \text{ for every } x \in R \setminus I.$$

Proof. *($\Rightarrow$)* Suppose I is maximal and $x \in R \setminus I$. Then $I \subsetneq (x, I)$ (since $x \in (x, I) \setminus I$), and (x, I) is an ideal. By maximality, $(x, I) = R$.

($\Leftarrow$) Suppose for contradiction I is not maximal. Then there exists an ideal J with $I \subsetneq J \subsetneq R$. Pick any $x \in J \setminus I$. Then $(x, I) \subseteq J \subsetneq R$, contradicting $(x, I) = R$. Hence I is maximal. □

2 The Quotient Characterisation

Theorem 2.1: Quotient test for maximal ideals (from lecture; cf. Lee, Thm. 9.20)

Let R be a nonzero commutative ring and I an ideal of R. Then

$$I \text{ is maximal} \iff R/I \text{ is a field.}$$

Proof. *($\Rightarrow$)* Suppose I is maximal, and let $x \in R \setminus I$. Since R is nonzero and commutative, it suffices to show $x + I$ is a unit in R/I (i.e. has a multiplicative inverse). By Lemma 1.2, $(x, I) = R$. So $\exists\, r \in R$ such that $xr + i = 1$ for some $i \in I$. Hence

$$(x + I)(r + I) = xr + I = (1 - i) + I = 1 + I.$$

So $r + I$ is the multiplicative inverse of $x + I$. Therefore R/I is a field.

($\Leftarrow$) Suppose R/I is a field and let $x \in R \setminus I$.
WTS: $(x, I) = R$. Then by Lemma 1.2, I is maximal.

Since $x + I$ is invertible in R/I, $\exists\, y \in R$ such that

$$(x + I)(y + I) = xy + I = 1 + I.$$

Hence $\exists\, i \in I$ with $xy + i = 1$, so $1 \in (x, I)$. Hence $R \subseteq (x, I)$.

The reverse inclusion is clear, so $R = (x, I)$. □

3 Maximal Implies Prime

Proposition 3.1: Maximal $\Rightarrow$ prime (from lecture; cf. Lee, Thm. 9.22)

Every maximal ideal of a nonzero commutative ring with identity is prime.

Proof.

$$\begin{aligned} I \text{ is maximal} &\iff R/I \text{ is a field} && \text{(Theorem 2.1)} \\ &\implies R/I \text{ is an integral domain} \\ &\iff I \text{ is prime} && \text{(Lecture 25, Theorem 2.1).} \end{aligned}$$

$\square$

Remark 3.1: The converse is NOT true (from lecture)

There exist prime ideals that are not maximal. Two key examples:

- $(x) \trianglelefteq \mathbb{R}[x,y]$ is prime but not maximal. Indeed, $\mathbb{R}[x,y]/(x) \cong \mathbb{R}[y]$, an integral domain but not a field.
- $(x) \trianglelefteq \mathbb{Z}[x]$ is prime but not maximal. By Example 3.1 of Lecture 25, $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, an integral domain but not a field.

4 Maximal Ideals in $\mathbb{C}[x]$

Example 4.1: $(x-\alpha)$ is maximal in $\mathbb{C}[x]$ (from lecture; cf. Lee, Example 9.30)

For every $\alpha \in \mathbb{C}$, the ideal $(x-\alpha) \trianglelefteq \mathbb{C}[x]$ is maximal (and hence also prime).

Define the evaluation-at-α map

$$\varphi_\alpha : \mathbb{C}[x] \to \mathbb{C}, \qquad f(x) \mapsto f(\alpha).$$

(1) φ_α is a ring homomorphism (evaluation at α).

(2) $\ker \varphi_\alpha = (x-\alpha)$.
($\supseteq$) For all $f(x) \in \mathbb{C}[x]$, $\varphi_\alpha\big((x-\alpha)f(x)\big) = (\alpha-\alpha)f(\alpha) = 0$, so $(x-\alpha) \subseteq \ker \varphi_\alpha$.

($\subseteq$) Let $f(x) \in \ker \varphi_\alpha$. Then $\varphi_\alpha(f(x)) = f(\alpha) = 0$, so α is a root of $f(x)$.

Recall the Fundamental Theorem of Algebra. For every $f(x) \in \mathbb{C}[x]$ with $\deg(f(x)) \geq 1$,

$$f(x) = a\prod_{i=1}^{n}(x-\alpha_i), \qquad a, \alpha_i \in \mathbb{C}.$$

Since α is a root, $(x-\alpha) \mid f(x)$. Hence $f(x) \in (x-\alpha)$.

Therefore $\ker \varphi_\alpha = (x-\alpha)$.

By the First Isomorphism Theorem,

$$\mathbb{C}[x]/(x-\alpha) \;\cong\; \mathbb{C}.$$

Since $\mathbb{C}$ is a field, $(x-\alpha)$ is maximal (and also prime) by Theorem 2.1. $\square$

Exercise 4.1: Maximal ideal in $\mathbb{C}[x, y]$ (from lecture)

For $\alpha, \beta \in \mathbb{C}$, the ideal $(x - \alpha, y - \beta) \trianglelefteq \mathbb{C}[x, y]$ is maximal.

Checklist of Skills

- ☐ State and prove the $(x, I) = R$ criterion for maximality.
- ☐ State and prove the quotient test: I maximal $\iff$ R/I is a field.
- ☐ Prove that every maximal ideal is prime, and explain why the converse fails (with examples in $\mathbb{R}[x, y]$ and $\mathbb{Z}[x]$).
- ☐ Verify that $(x - \alpha) \trianglelefteq \mathbb{C}[x]$ is maximal using the evaluation map and the Fundamental Theorem of Algebra.
- ☐ (Exercise) Verify that $(x - \alpha, y - \beta) \trianglelefteq \mathbb{C}[x, y]$ is maximal.